PROJECT STATUS REPORT

ClaudeCode Advancements

Research, tools, and systems for AI-assisted development

8,959	158,077	9	930
TESTS PASSING	LINES OF CODE	MODULES	COMMITS

37	398	18	33d
master tasks	intelligence findings	live hooks	project age

Session 156 · 2026-03-24

github.com/mpshields96/ClaudeCodeAdvancements

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EXECUTIVE SUMMARY

Executive Summary

ClaudeCodeAdvancements is a research and development project building validated tools, hooks, and systems for Claude Code users and AI-assisted development workflows. Launched 2026-02-19 with 156 development sessions completed. All five original research frontiers are production-complete with 8,959 automated tests across 9 modules. The project has expanded into 37 master-level tasks covering autonomous intelligence gathering, self-learning loops, academic research integration, and professional design output. Zero external Python dependencies.

SINCE LAST REPORT — Session 156 → 156

+71	+29	+3
Tests	LOC	Commits

Project Health

TESTS

8,959 / 8,959

223 suites — 100% pass rate

CODEBASE

158,077

Source: 64,085 Test: 93,992

Test-to-source ratio: 1.47:1

VELOCITY

930 commits

28.2/day · 6/session

33 days · 156 sessions

MODULES

9

160 source files · 223 test files

0 stubs · 0 syntax errors

MASTER TASKS

37

8 done · 23 active · 5 pending

INTELLIGENCE

398 findings

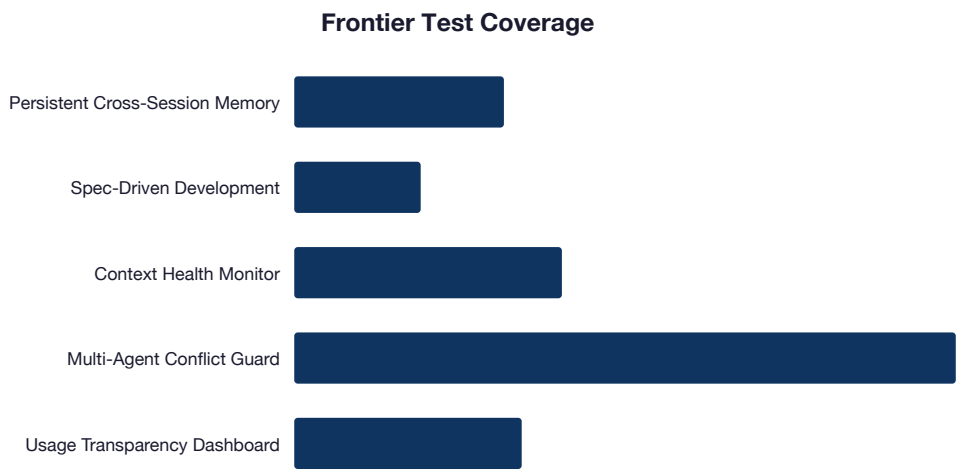
1242 papers · 18 live hooks

Zero external dependencies

FIVE FRONTIERS

Five Frontiers

The five core research areas that drive CCA's development, each validated through community intelligence, Anthropic research, and developer surveys.



1 Persistent Cross-Session Memory

Active CRITICAL

Every session starts from zero — this gives Claude a brain that persists.

TESTS: **340** LOC: **1,758**

2 Spec-Driven Development

Active HIGH

Unstructured prompting produces poor architecture. Enforce requirements > design > tasks > implement.

TESTS: **205** LOC: **1,191**

3 Context Health Monitor

Active HIGH

Context rot causes silent output degradation. Monitor, warn, and auto-handoff.

TESTS: **434** LOC: **2,871**

4 Multi-Agent Conflict Guard

Active HIGH

Parallel agents overwrite each other. Guard credentials, paths, and system modifications.

TESTS: **1073** LOC: **6,196**

5 Usage Transparency Dashboard

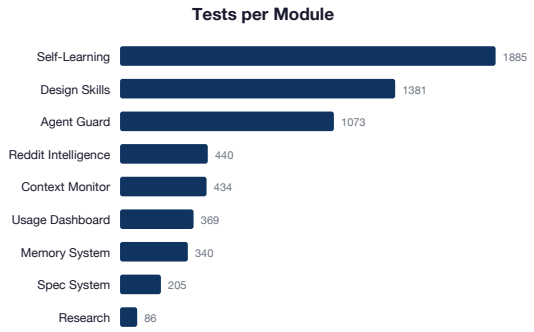
Active MEDIUM

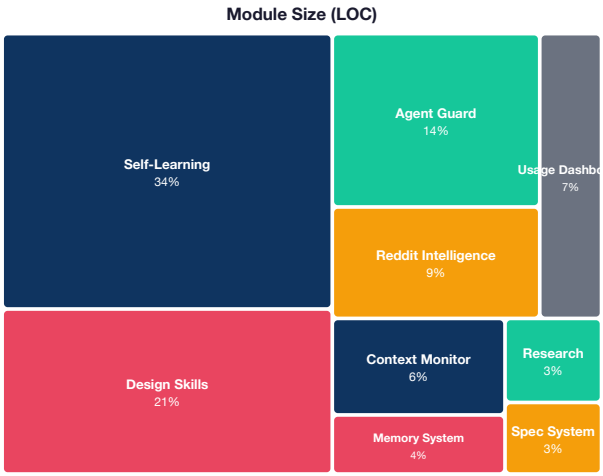
No real-time token/cost visibility. Counter, alerts, and structural completeness.

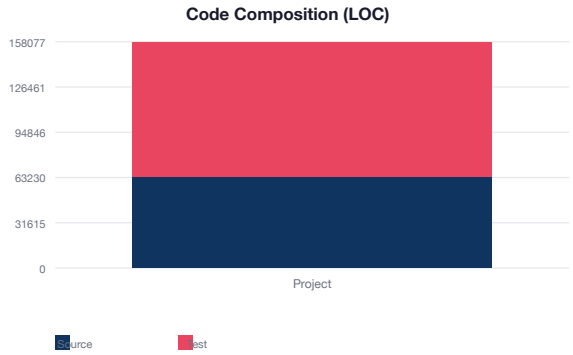
TESTS: **369** LOC: **3,030**

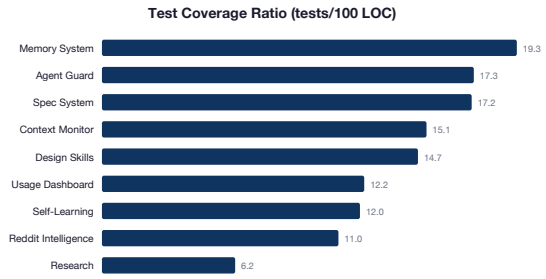
MODULE DEEP-DIVES

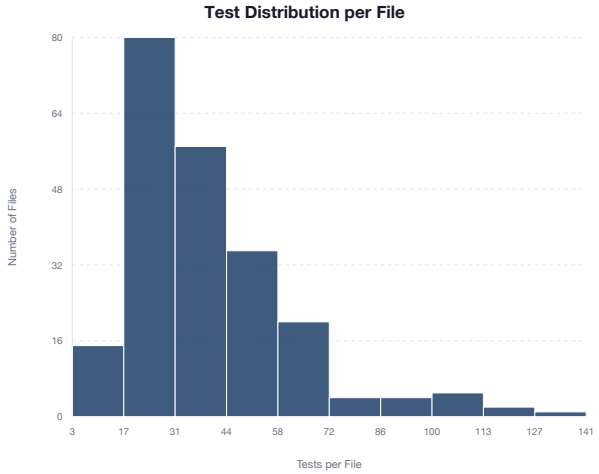
Module Deep-Dives











Memory System

Active

TESTS **340** LOC **1,758** FILES **4** `memory-system/`

Persistent cross-session memory. Captures decisions, preferences, and context across Claude Code sessions using hooks and MCP.

KEY COMPONENTS

- Capture hook (PostToolUse + Stop)
- CLI viewer (stats, search, list)
- 8-char hex ID schema
- MCP retrieval server
- Compaction-resistant handoff

WHY THIS MATTERS Means Claude remembers your preferences and past decisions across sessions — no more re-explaining context every time you start a new chat.

Spec System

Active

TESTS **205** LOC **1,191** FILES **4** `spec-system/`

Spec-driven development workflow. Enforces requirements > design > tasks > implement sequence with approval gates.

KEY COMPONENTS

- /spec:requirements interview scaffold
- /spec:tasks atomic decomposer
- PreToolUse spec validator hook
- /spec:design architecture generator
- /spec:design-review (4 expert personas)
- UserPromptSubmit auto-activation

WHY THIS MATTERS Forces Claude to plan before coding. You get a reviewable blueprint (requirements, design, tasks) instead of spaghetti code from vague prompts.

Context Monitor

Active

TESTS **434** LOC **2,871** FILES **10** `context-monitor/`

Context health monitoring with adaptive thresholds. Tracks token usage, warns at quality ceilings, blocks exit at critical levels.

KEY COMPONENTS

- PostToolUse token meter
- Stop hook auto-handoff
- Automatic session wrap trigger
- PreToolUse threshold alert
- Compact anchor (every 10 turns)
- Session pacer (2-3h autonomous runs)

WHY THIS MATTERS Warns you when Claude's memory is getting full so it doesn't silently start forgetting instructions mid-session. Prevents the 'why did it ignore my rules?' problem.

Agent Guard

Active

TESTS **1073** LOC **6,196** FILES **21** `agent-guard/`

Multi-agent safety, conflict prevention, and Senior Dev code review. Guards credentials, network ports, dangerous paths, and provides automated quality scoring on every write.

KEY COMPONENTS

- Mobile approver (ntfy.sh push)
- Network/port exposure guard
- Path validator (LIVE in hooks)
- File ownership manifest
- Bash command safety guard (LIVE in hooks)
- PR effort scorer (1-5 scale)
- False positive filter
- Tech debt tracker (trend analysis)
- Senior Dev orchestrator hook (LIVE)
- Credential extraction guard
- Content scanner (9 threat categories)
- Session guard (slop detection)
- Edit retry guard (LIVE in hooks)
- SATD detector (TODO/FIXME/HACK markers)
- Code quality scorer (0-100, A-F)
- Review classifier (6 categories)
- ADR reader (architecture decision discovery)

WHY THIS MATTERS Stops Claude from accidentally deleting files, exposing API keys, or breaking your computer when running autonomously overnight.

Usage Dashboard

Active

TESTS **369** LOC **3,030** FILES **7** `usage-dashboard/`

Token and cost transparency. CLI counter, OpenTelemetry receiver, cost alerts, and structural completeness checker.

KEY COMPONENTS

- Token counter CLI
- PreToolUse cost alert hook
- OTLP HTTP/JSON receiver
- /arewedone completeness checker

WHY THIS MATTERS Shows you exactly how many tokens and dollars each session costs, so you can spot waste and stay within your Max subscription limits.

Reddit Intelligence

Complete

TESTS **440** LOC **3,983** FILES **7** `reddit-intelligence/`

Community signal research. Discovers tools, patterns, and pain points from Reddit and GitHub to drive development priorities.

KEY COMPONENTS

- Reddit reader (posts + all comments)
- GitHub repo evaluator
- Subreddit profile registry (10 profiles)
- Autonomous scan pipeline
- Sandboxed repo tester

WHY THIS MATTERS Automatically finds the best new tools, tips, and techniques from Reddit so you don't have to scroll through hundreds of posts yourself.

Self-Learning

Complete

TESTS **1885** LOC **15,666** FILES **39** `self-learning/`

Cross-session improvement through structured observation, pattern detection, autonomous proposals, and research ROI tracking.

KEY COMPONENTS

- JSONL event journal
- Trace analyzer (4 detectors)
- Sentinel adaptive mutation
- Findings resurfacer
- Overnight detector (time-stratified analysis)
- Trade reflector (5 pattern detectors)
- Pattern reflector + strategy engine
- YoYo improvement loop
- Academic paper scanner
- Skillbook injection hook (LIVE)
- Research outcomes ROI tracker

WHY THIS MATTERS Makes your Kalshi bot smarter over time by analyzing what worked and what didn't. The bot learns from its own wins and losses instead of repeating mistakes.

Design Skills

Active

TESTS **1381** LOC **9,387** FILES **14** `design-skills/`

Professional visual output. Typst-based PDF reports, slides, HTML dashboards, SVG charts, website generation, and daily project snapshots.

KEY COMPONENTS

- Report generator (Typst pipeline)
- Dashboard generator (HTML)
- Website generator (landing + docs)
- Design guide + visual language
- Slide generator (16:9 PDF)
- Chart generator (SVG)
- Daily snapshot engine (metric diffs)

WHY THIS MATTERS Generates professional PDF reports and charts automatically — no manual formatting. One command gives you a polished status overview you could show anyone.

Research

Complete

TESTS **86** LOC **1,397** FILES **3** `research/`

R&D tools and experimental capabilities including iOS project generation and Xcode build wrapper.

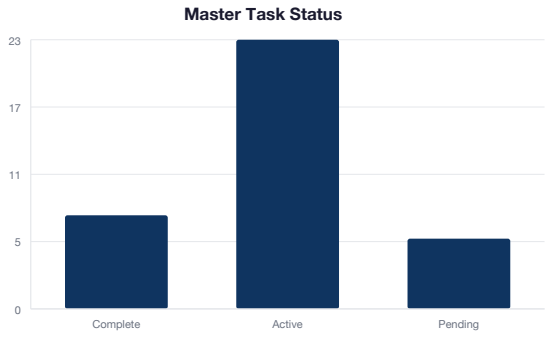
KEY COMPONENTS

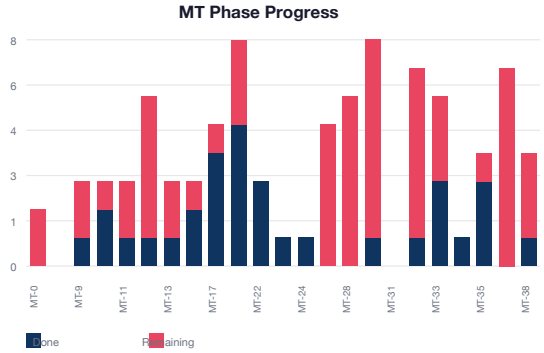
- iOS project generator (SwiftUI)
- Xcode build wrapper

WHY THIS MATTERS R&D sandbox for experimental features. Things get prototyped here before becoming real modules — keeps the main codebase clean.

MASTER TASKS

Master Tasks





37 tasks defined · **8 complete** · **23 active** · 5 pending · 1 blocked

Completed

8 tasks delivered

MT-2	**Mermaid Architecture Diagrams in Spec System**		Complete
MT-3	**Virtual Design Team Plugin**		Complete
MT-4	**Frontend Design Excellence via Design Vocabulary**		Complete
MT-6	**On-Demand Subreddit Scanner**	✓ 43 tests	Complete
MT-7	**Programmatic Trace Analysis for Self-Learning**	✓ 50 tests	Complete
MT-20	**Senior Developer Agent**	✓ 890 tests	Complete
MT-26	**Financial Intelligence Engine**	✓ 43 tests	Complete
MT-29	**Cowork + Claude Pro Bridge Hivemind**		Complete

Active

23 tasks in progress

MT-0 Kalshi Bot Self-Learning Integration

✓ 24 tests **In Progress**

Phase 0/2

COMPLETE (Phase 1: S21, Phase 2: deployed via polymarket-bot monitoring chat).

MT-1 Maestro-Style Visual Grid UI

In Progress

MOSTLY SELF-RESOLVED (S96). Multiple tools now exist:

MT-9 Autonomous Cross-Subreddit Intelligence Gathering

✓ 54 tests **In Progress**

Phase 1/3

Phase 2 COMPLETE (Session 31). Delivered:

MT-10 YoYo-Pattern Continuous Self-Learning and Self-Building

✓ 64 tests **In Progress**

Phase 2/3

Phase 1 COMPLETE (Session 28), QualityGate added (Session 29), E2E validated (Session 29).

MT-11 Autonomous GitHub Repository Intelligence

✓ 62 tests **In Progress**

Phase 1/3

Phase 2 COMPLETE (Session 83). Delivered:

MT-12 Academic Research Paper Integration

✓ 89 tests **In Progress**

Phase 1/6

Phase 4 COMPLETE (Session 101). Delivered:

MT-13 iOS App Development Capability

✓ 57 tests **In Progress**

Phase 1/3

Phase 2 COMPLETE (Session 49). Delivered:

MT-14 Autonomous Re-Scanning of Previously Scanned Subreddits

In Progress

Phase 2/3

Phase 1 COMPLETE (Session 35). Delivered:

MT-17 UI/Design Excellence and Professional Report Generation

✓ 42 tests **In Progress**

Phase 4/5

Phase 1 COMPLETE (Session 41). Delivered:

MT-21 Hivemind Multi-Chat Coordination

✓ 69 tests **In Progress**

Phase 5/8

Infrastructure COMPLETE. Phase 1 validation IN PROGRESS (S86). **Next: 1-3 more validation tests, then Phase 1 gate assessment.**

MT-22 Autonomous 1-Hour Loop

✓ 66 tests **In Progress**

Phase 3/3

GATE PASSED (S99) for CLI auto-loop. Now pivoting to DESKTOP ELECTRON APP automation (Matthew S130/S132 directive — #1 priority, equal to YoYo).

MT-23 Mobile Remote Control v2

In Progress

EXTERNALLY RESOLVED (S112). Claude Code Channels shipped March 20, 2026. **Next: Install Telegram channel plugin and test with Kalshi bot session.**

MT-24 Visualization & Graphics Engine

✓ 41 tests **In Progress**

Phase 1/1

Phase 1 COMPLETE (S111) — `trading_chart.py` (5 SVG chart types: PnL curve, win rate, strategy matrix, drawdown, heatmap). 41 tests.

MT-27 CCA Nuclear v2

✓ 27 tests **In Progress**

Phase 0/5

Phases 1-5 COMPLETE.

MT-28 Self-Learning v2

✓ 73 tests **In Progress**

Phase 0/6

COMPLETED (S111). All 6 phases done:

MT-30 Session Daemon

✓ 116 tests **In Progress**

Phase 1/8

Phase 8 COMPLETE (S128). S129: +preflight command, +rich --status with audit log, +AUTOLOOP_SETUP.md. Ready for live supervised dry run.

MT-31 Gemini Pro Integration

In Progress

S125 E2E validation: Gemini Flash WORKS via @rlabs-inc/gemini-mcp (v0.8.1). Pro UNAVAILABLE — Google One gives web app Pro access only, not API access. API Pro requires separate Google Cloud billing. Matthew directive: Flash only (Option 1). Scope narrowed to Gemini Flash integration.

MT–32 **Visual Excellence & Design Engineering** ✓ 890 tests In Progress

Phase 1/7

NEW (S118). Phase 1 starting — report_charts.py integration + research scan.

MT–33 **Strategic Intelligence Report** In Progress

Phase 3/6

Unknown

MT–34 **Medical AI Tool — OpenEvidence Replacement** In Progress

IDEA LOGGED — Matthew needs more time to refine the concept. Do NOT start building until Matthew gives the green light.

MT–35 **Background Autoloop — Non-Intrusive Desktop Loop** ✓ 23 tests In Progress

Phase 3/4

- Phase 1 COMPLETE (S142): save/restore frontmost app around autoloop trigger (~3-5s takeover)

MT–37 **AI-Driven Investment Research & Portfolio Intelligence** In Progress

Phase 0/7

IDEA LOGGED (S148) — Long-term, research-first. Do NOT start building until Phase 1 research is complete. No rush — Matthew directive.

MT–38 **Peak/Off-Peak Token Budget System** In Progress

Phase 1/4

Phase 1 COMPLETE (S154). Rule active globally. Phase 2-4 are enhancement.

Pending & Blocked

6 tasks awaiting start or unblocked

MT–5 **Claude Pro ↔ Claude Code Bridge** Not Started

MT–8 **iPhone Remote Control Perfection** Not Started

MT–18 **Academic Writing Workspace** Not Started

MT–19 **Local LLM Fine-Tuning Capability** Not Started

MT-25	**Presentation Generator**	Blocked
MT-36	**Session Efficiency Optimizer — Quality-Preserving Speed**	Not Started

PRIORITY QUEUE

Priority Queue

Active tasks ranked by decay-based priority scoring. Score = base value + aging penalty. Higher = work on this first.

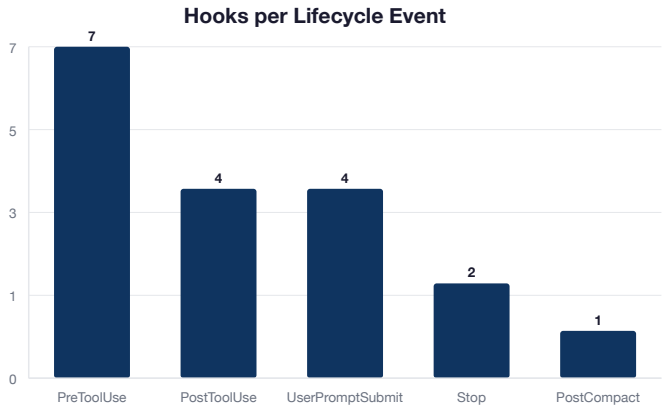
1	MT-21	**Hivemind coordination** Phase 2 PASSED (6th). Phase 3 = 3-chat	10
2	MT-33	**Strategic Intelligence Report** Phase 4-6: chart integration, self-reference, hardening. kalshi_data_collector + learning_data_collector + report_differ built (S122-S123).	10
3	MT-23	**Mobile Remote Control v2** Research: evaluate Telegram/Discord channels MCP	9
4	MT-26	**Financial Intelligence Engine** CCA scope COMPLETE. Tier 3 Phase 2 deferred (needs numpy). 79 pipeline tests.	8
5	MT-32	**Visual Excellence & Design Engineering** Phase 2+: nuclear scan, Kalshi charts, consistency. report_charts wired (S117).	8
—	MT-34	**Medical AI (OpenEvidence replacement)** BLOCKED: Matthew refining concept. Do not start.	6
8	MT-29	**Cowork + Pro Bridge Hivemind** Research: evaluate Cowork, bridge Pro↔Code	5
9	MT-18	**Academic writing** Research: install/evaluate ClaudePrism	7
10	MT-13	**iOS/macOS app dev** Phase 3: first real app	7

LIVE INFRASTRUCTURE

Live Infrastructure

18 hooks integrated into Claude Code via settings.local.json. Every tool call passes through this pipeline.

Event	Matcher	Hook	Purpose
`PostToolUse`	*	meter.py	Token counter
`PostToolUse`	*	compact_anchor.py	Context anchor writes
`PostToolUse`	*	senior_dev_hook.py	SATD + effort + quality scoring
`PostToolUse`	*	queue_hook.py	Cross-chat queue context
`PreToolUse`	*	alert.py	Warn/block at red/critical
`PreToolUse`	*	cost_alert.py	Cost threshold warning
`PreToolUse`	*	path_validator.py	Dangerous path detection
`PreToolUse`	*	edit_guard.py	Edit retry prevention
`PreToolUse`	*	validate.py	Spec guard + plan compliance
`PreToolUse`	Bash	credential_guard.py	Credential extraction guard
`PreToolUse`	Bash	bash_guard.py	Command safety (network, packages, system)
`UserPromptSubmit`	*	skill_activator.py	Spec auto-activation
`UserPromptSubmit`	*	skillbook_inject.py	Strategy injection
`UserPromptSubmit`	*	capture_hook.py	Real-time memory capture
`UserPromptSubmit`	*	queue_hook.py	Cross-chat queue injection
`Stop`	*	auto_handoff.py	Block exit at critical
`Stop`	*	capture_hook.py	Session-end memory capture
`PostCompact`	*	post_compact.py	Recovery + journal logging



INTELLIGENCE & RESEARCH

Intelligence & Research

Reddit Intelligence

FINDINGS BY VERDICT

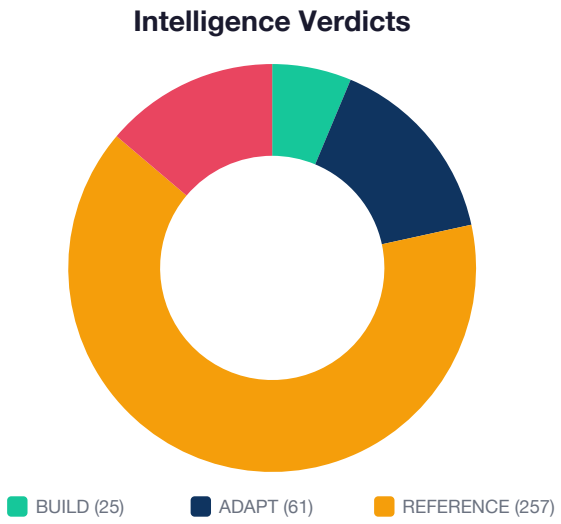
BUILD	25
ADAPT	61
REFERENCE	257
PERSONAL	0
SKIP	55

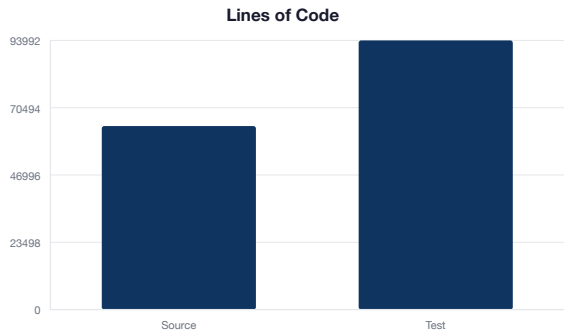
7 subreddits · 30 repos evaluated

Self-Learning System

Principles	66 active (avg score: 0.5)
Journal Sessions	109 tracked
Wins / Pains	72 / 53
Papers Logged	**1242**
Sentinel Rate	0%
Research ROI	**46 deliveries, 0 profitable**

YoYo loop: observe → detect → hypothesize → build → validate

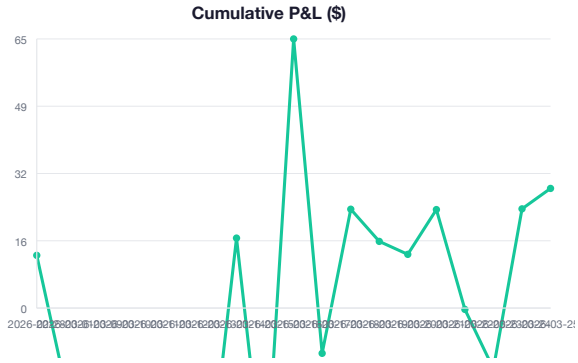

Intelligence Verdicts
BUILD (25)
ADAPT (61)
REFERENCE (257)

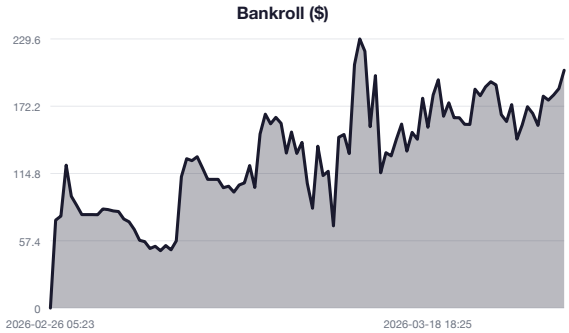

Lines of Code
93992
70494
46996
23498
0
Source
Test

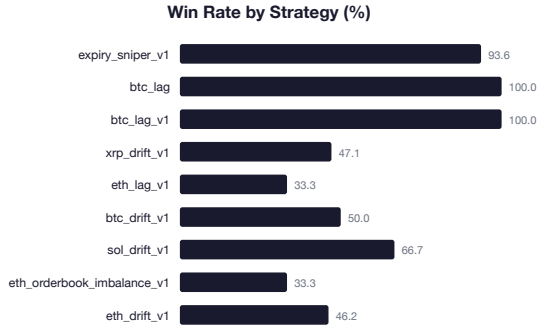
FINANCIAL ANALYTICS — KALSHI BOT

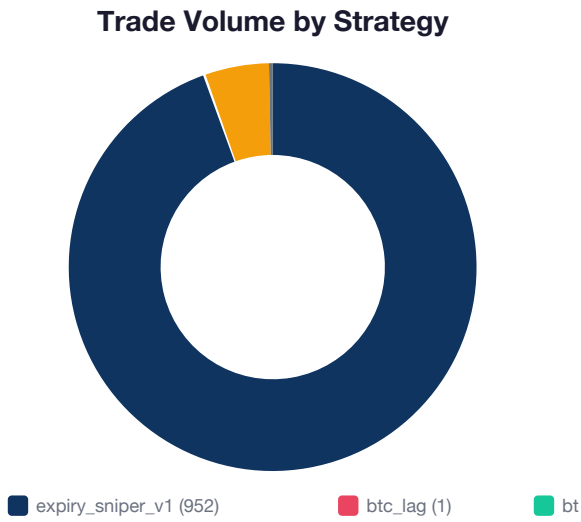
Financial Analytics — Kalshi Bot

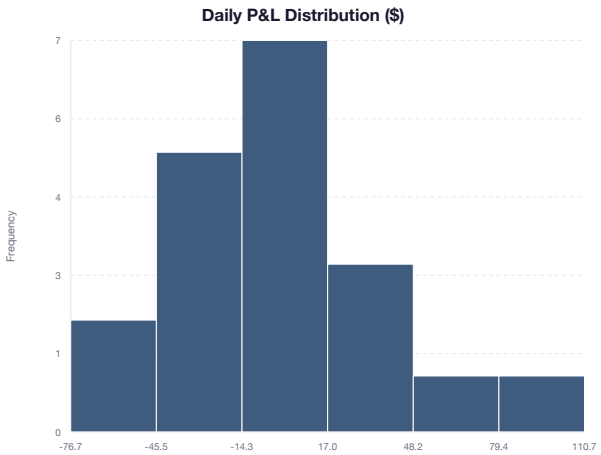
LIVE TRADES	WIN RATE	TOTAL P&L	SETTLED
1306	81.6%	$29.2	1305

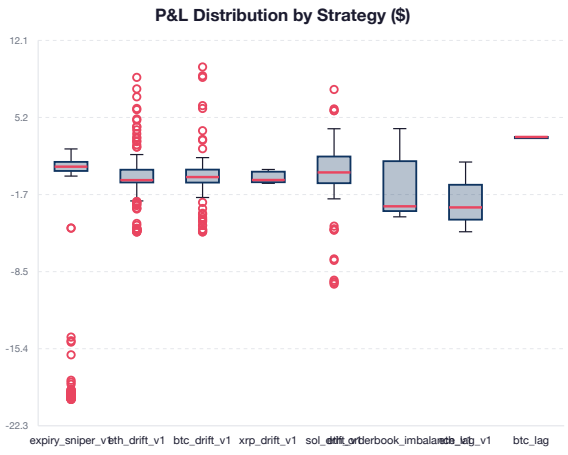












Strategy Performance

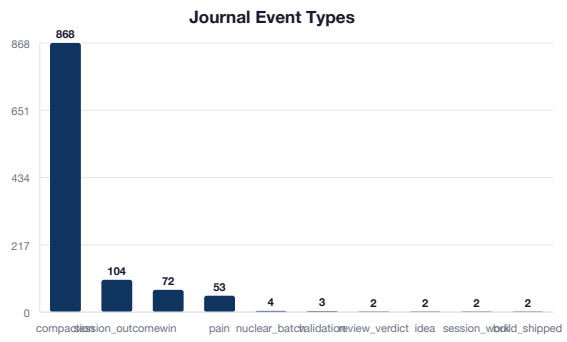
Strategy	Trades	Wins	Win %	Avg P&L	Total P&L
expiry_sniper_v1	952	890	93.6%	$0.11	**$104.62**
btc_lag	1	1	100%	$3.5	**$3.5**
btc_lag_v1	1	1	100%	$0.57	**$0.57**
xrp_drift_v1	51	24	47.1%	$−0.07	**$−3.8**
eth_lag_v1	3	1	33.3%	$−2.18	**$−6.53**
btc_drift_v1	80	40	50%	$−0.12	**$−9.53**
sol_drift_v1	45	30	66.7%	$−0.31	**$−14.08**
eth_orderbook_imbalance_v1	15	5	33.3%	$−1.21	**$−18.2**

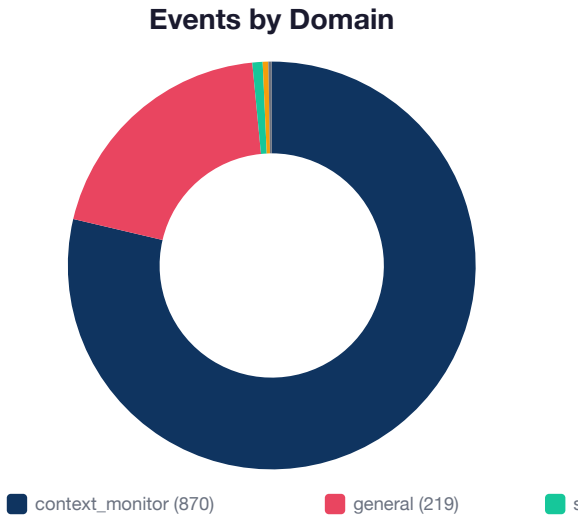
eth_drift_v1	158	73	46.2%	$−0.17	**$−27.35**

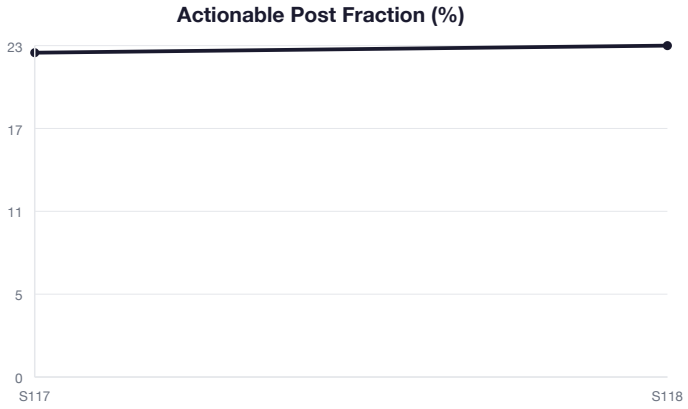
SELF-LEARNING INTELLIGENCE

Self-Learning Intelligence

JOURNAL ENTRIES	WINS / PAINS	CURRENT APF	POSTS REVIEWED
1121	72 / 53	23.2%	340







ARCHITECTURE DECISIONS

Architecture Decisions

Decision	Rationale
Local-first storage	User owns all data, no cloud dependency
Stdlib-first (no pip)	Zero dependency management overhead
Hook-based delivery	Native Claude Code integration points
Adaptive context thresholds	Fixed percentages break at 1M windows
JSONL for all logging	Append-only, grep-friendly, no corruption risk
QualityGate geometric mean	Prevents Goodhart's Law gaming of metrics
Whitelist-first phishing detection	Zero false positives on legitimate domains
Typst over WeasyPrint	Single binary, millisecond compile, JSON-native

RISKS & BLOCKERS

Risks & Blockers

MT-25 Presentation Generator (Matthew's Style)

BLOCKER

NOT STARTED. BLOCKED on style samples from Matthew.

2 MTs stagnating

RISK

2 active tasks have negative stagnation scores — no progress in recent sessions.

Mitigation: Review priority picker output. Consider deprioritizing or archiving stalled work.

5 severity-3 learnings active

DEBT

High-severity patterns requiring ongoing vigilance across all sessions.

Mitigation: Encoded in global rules. Auto-enforced by hooks where possible.

NEXT PRIORITIES

Next Priorities

1. **Optimize wrap time**
   Matthew flagged wrap is too long. Reduce overhead without quality loss
2. **Continue Kalshi cross-chat coordination**
   check for new requests
3. **K2 follow-up**
   assist with gdp_sniper.py code review if built
4. **Read TODAYS_TASKS.md for full task list**

HONEST ASSESSMENT

Honest Assessment

Objective gaps, limitations, and areas where the project falls short of its stated goals. Included for accountability.

11 master tasks stalled at Phase 1

GAP

MT-0, MT-9, MT-11, MT-12, MT-13, MT-27, MT-28, MT-30, MT-32, MT-37, MT-38 — started but no subsequent phases shipped. Risk of scope sprawl without depth.

Single-developer project with no external users

LIMITATION

All tooling built for one developer. No community testing, no feedback loop from other Claude Code users. Limits validation of 'community-demanded' features.

High test count (6,213) may overstate coverage

NUANCE

Average 57 tests per source file. Many test trivial properties (dict keys, return types). Integration test coverage across modules is limited to 1 suite.

1 task(s) blocked with no resolution timeline

BLOCKER

MT-25: Presentation Generator

No research outcome tracking file found

GAP

~/.cca-research-outcomes.jsonl missing. CCA cannot measure which deliveries to Kalshi produced profit.

CI/CD pipeline exists but unverified

NUANCE

GitHub Actions workflow defined but no evidence of runs in git history. All testing is local-only.

STATUS

All Systems Operational

8,959	158,077	156	930
TESTS	LOC	SESSIONS	COMMITS

9 modules · 37 master tasks · 223 test suites · Zero external dependencies

8	23	398
MTs Complete	MTs Active	Findings

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